Lab 7: Fuzz Testing - Weather Alert Classifier

Student ID: PES2UG24CS593 | Repository: L6_L7_32_weather_alert

Hypothesis property testing and defensive input validation

Starter app | python weather_alert.py

```
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): Temperature and Humidity: (35.5, 80)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): Wind Category: advisory

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): Rainfall Spread: 8.0
```

1. Starter application run

The assigned starter script ran with 35.5/80, 45, and 2,5.5,10. Its ordinary path worked before fuzzing.

2. Fuzz tests expose failures

2. Fuzz tests expose failures

Hypothesis run | starter implementation

```
$ python -m pytest -q --tb=line test_weather_alert.py
F.F.FF.                                                                  [100%]
FAIL test_parse_reading_handles_arbitrary_input: Hypothesis minimized malformed readings ("/", "", None)
FAIL test_wind_category_handles_arbitrary_input: invalid integer inputs ("", None)
FAIL test_wind_category_rejects_extremely_long_integer_text: 5000-digit text raises ValueError
FAIL test_rainfall_spread_handles_arbitrary_input: NaN, empty text, and None were not handled safely
4 failed, 3 passed in 5.85s
```

2. Fuzz tests expose failures

The original implementation failed 4 of 7 checks. Hypothesis found malformed values, non-finite rainfall, and an oversized integer conversion failure.

3. Hardened application run

3. Hardened application run

Fixed app | python weather_alert.py

```
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): Temperature and Humidity: (35.5, 80)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): Wind Category: advisory

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): Rainfall Spread: 8.0
```

3. Hardened application run

After validation was added, the same sample inputs still produce (35.5, 80), advisory, and 8.0 mm spread.

4. Re-run after the fix

4. Re-run after the fix

```
Hypothesis run | fixed implementation

.......                                                  [100%]
7 passed in 3.13s
```

4. Re-run after the fix

All 7 tests pass. Invalid readings are rejected safely; finite, nonnegative rainfall and the original wind thresholds remain covered.